

Spatiotemporal Evolution of Coupling Coordination between Artificial Intelligence and Technology Finance Development and Its Influencing Factors

Huan Chen¹, Muhan Liang^{2*}

¹ School of International Business, Hainan University, Haikou, China

² School of Information and Communication Engineering, Hainan University, Haikou, China

Abstract: Artificial intelligence (AI) deeply drives the quality improvement and efficiency enhancement of technology finance, while the deepening and expansion of technology finance also promote the iteration and application innovation of AI technology. Based on an analysis of the intrinsic coupling mechanism between AI and technology finance, and using provincial panel data of China from 2014 to 2023, this study comprehensively applies the coupling coordination degree model, the modified gravity model, and social network analysis to construct a spatial correlation network of coupling coordination. It further explores the evolutionary characteristics of the spatial correlation network in the coupling coordination relationship between AI and technology finance. The QAP regression model is employed to deeply analyze the driving factors of the spatial correlation network of coupling coordination. The findings reveal that: (1) The coupling coordination level between AI and technology finance development in China shows a trend of minor fluctuations, with significant regional gradient differences and dynamic evolution characteristics. The spatial correlation intensity exhibits a distribution pattern of “dense in the east and loose in the central and western regions.” (2) The spatial correlation network of coupling coordination displays stable and complex structural characteristics, with overall connectivity and stability continuously strengthening and the network structure gradually optimizing. From the perspective of individual network characteristics, provinces such as Beijing, Shanghai, and Jiangsu have long occupied core positions in the network, while most central and western provinces remain on the periphery. However, the network as a whole shows a flattening development trend, with connection paths becoming increasingly diversified. (3) The spatial correlation network can be divided into four plates: net beneficiary, broker, bidirectional spillover, and net spillover. Cross-plate associations dominate the interactions between these plates, with significant spillover effects observed. (4) QAP regression analysis indicates that geographical proximity, differences in innovation levels, disparities in technology market development, variations in economic development levels, and differences in urbanization levels are important factors driving the formation and evolution of the spatial correlation network. This research provides theoretical insights and policy implications for promoting the deep integration of AI and technology finance and achieving high-level coordinated development.

Keywords: Artificial Intelligence; Technology Finance; Coupling Coordination; Spatial Correlation Network; Social Network Analysis; QAP Regression Model

*Author Correspondence: (First author) Huan Chen (chenhuan9422@163.com); (Corresponding author) Muhan Liang (479332290@qq.com).
Received 15 February 2026, Revised 01 March 2026, Accepted 15 March 2026

1 Introduction

Artificial intelligence is the core driving force of the new round of scientific and technological revolution and industrial transformation. Technology finance refers collectively to the financial systems, policies, instruments, products, and related financial activities that promote technological innovation, the commercialization of outcomes, and the development of technology industries (Zou et al., 2024). It serves as a key financial form supporting industrial upgrading. Together, artificial intelligence and technology finance constitute a vital engine for high-quality development in the digital economy era. With its powerful capabilities in data-driven analysis, intelligent decision-making, and risk identification (Danielsson et al., 2022), artificial intelligence is profoundly reshaping the service models, operational processes, and risk-pricing mechanisms of technology finance. Simultaneously, the application scenarios and sustained financing demands within technology finance provide a crucial testing ground and market traction for the iterative advancement and commercial implementation of artificial intelligence technologies. Currently, China's technology finance system still faces practical challenges such as insufficient service precision and high risk management costs. There is an urgent need for artificial intelligence to inject new momentum. Meanwhile, the development of the artificial intelligence industry itself relies on continuous support from multi-level capital markets and efficient financial services. As artificial intelligence technology enters a phase of deeper application and technology finance advances toward a new cycle of high-quality development, promoting the deep integration and coordinated progress of the two has become an inevitable path and a strategic consensus. This is essential for enhancing the overall efficiency of the national scientific and technological innovation system and strengthening the financial security barrier.

Reviewing the existing literature, studies on technology finance have primarily been conducted at the macro, micro, and policy evaluation levels. At the macro level, scholars have focused on the impact mechanisms of technology finance on regional capital (Sheng et al., 2025), new-quality productive forces (Zou et al., 2024), and other aspects, as well as research on its coupling coordination with the high-quality development of the real economy (Liu & Liu, 2022). At the micro level, studies have centered on the effects of technology finance on firms' total factor productivity (TFP) (Wang & Liu, 2025), radical green innovation efficiency (Li et al., 2025), and labor employment (Zhou, 2024). In addition, some scholars have provided abundant evidence for understanding the operational mechanism and implementation effect of technology finance from perspectives such as the integration of the scientific and technological innovation chain (Li et al., 2022) and pilot policy evaluation (Guo, 2024; Ma, 2019). Furthermore, the credit support mechanism of technology finance has also attracted attention (She et al., 2025), and its association with breakthroughs in key digital technologies has been explored (Zhou et al., 2025).

Regarding studies on the economic effects of AI, specifically, in terms of theoretical discussions, scholars such as Pan et al. (2025) and Fan et al. (2025) have explored the theoretical essence and practical orientation of AI empowerment. At the micro level, some scholars have focused on the impacts of AI on firm resilience (Yang et al., 2025), firm value (Lui et al., 2022), corporate growth (Babina et al., 2024), operational efficiency (Mishra et al., 2022), innovation performance (Lin et al., 2025), and total factor productivity (Czarnitzki et al., 2023), as well as the pathway of influence on employee performance (Wei & Yang, 2025). In the financial field, some scholars have discussed AI empowering the high-quality development of securities firms (Wang & Zhao, 2025), its relationship with high-quality financial development (Jiang, 2025.), and the resulting financial opportunities and ethical challenges (Bonaparte, 2024; Ma & Han, 2025).

It is evident that current research on artificial intelligence and technology finance is gradually expanding and deepening. However, studies predominantly focus on unidirectional enabling effects or qualitative analyses of single dimensions. Few works systematically explore the spatial network correlations in their coupling coordination, and insufficient attention has been paid to the spatiotemporal evolution of their coupling coordination levels over time, as well as the development trends of their spatial correlation networks. Furthermore, discussions on the driving factors behind the spatial correlations in AI-technology finance coupling coordination remain scarce, leaving the internal mechanisms and transmission pathways of network formation unclear. In reality, the coupled development of AI and technology finance is not an isolated regional phenomenon. Rather, it involves close interregional flows and interactive linkages of technology and capital elements across space. If the networked

connections among geographical units from a spatial correlation perspective are neglected, it becomes difficult to comprehensively grasp the characteristics of cross-regional factor flows, spatial spillover effects, and coordination mechanisms during their coupling coordination process. Therefore, investigating the spatiotemporal evolution patterns and influencing factors of the coupling coordination between AI and technology finance development serves both as a practical necessity for addressing bottlenecks in financing technological innovation and improving the quality and efficiency of financial services in supporting the real economy. It also represents a crucial pathway for promoting regional innovation synergy and facilitating high-quality economic development, making it a core issue in both theoretical research and practical exploration.

In summary, this study constructs a spatial correlation network of interprovincial coupling coordination between AI and technology finance development based on a modified gravity model. It employs social network analysis to deconstruct the characteristics and dynamic evolution patterns of its spatial correlation structure, thereby aiming to expand research on the synergistic development of AI and technology finance and provide scientific guidance for their deeper integration in practice. Furthermore, by introducing the QAP regression method, this paper empirically identifies the core driving factors of the spatial correlation in their coupling coordination, revealing the formation mechanisms and transmission pathways of the spatial network. Thus, it provides empirical support for refining the theoretical framework of AI-technology finance synergy.

The marginal contributions of this paper are primarily reflected in the following four aspects. First, in terms of research perspective, this study adopts the dual lenses of "coupling coordination" and "spatial correlation" to systematically analyze the spatiotemporal evolution characteristics of the coupling coordination between AI and technology finance development. This approach addresses the gap in existing research, which often focuses on unidirectional enabling effects and pays insufficient attention to cross-regional synergistic interactions. Second, regarding methodological innovation, this paper integrates social network analysis with a modified gravity model to more accurately characterize the intensity of spatial correlations in interprovincial coupling coordination. Based on this, it provides an in-depth revelation of the dynamic evolutionary patterns within the associated network structure. Third, from the perspective of mechanism analysis, the QAP regression model is employed, effectively avoiding the multicollinearity issues inherent in traditional econometric methods. This allows for the precise identification of key driving factors behind the spatial correlation network of coupling coordination and elucidates its spatial transmission mechanisms. The findings of this study can provide a scientific basis for the national-level coordination of industrial layouts for AI and technology finance, the optimization of science and technology innovation resource allocation, and the promotion of regional coordinated development. Furthermore, it contributes to facilitating the efficient flow of technology and capital factors, as well as fostering rational regional specialization. Ultimately, it offers decision-making support for constructing a nationwide synergistic ecosystem for AI and technology finance, thereby empowering high-quality development of the digital economy.

2 The mechanism of coupling coordination between AI and technology finance development

2.1 The impact pathways and effects of AI on technology finance development

As the core technological support in the era of the digital economy, AI exerts an impact on the development quality and effectiveness of technology finance through pathways such as reconstructing service models, optimizing risk management and control, and improving allocation efficiency.

By digitizing services, AI technology greatly shortens the business processing cycle of technology finance, reduces operational costs, and thereby improves the efficiency of technology finance services. In the link of financing matching, it can effectively match the financing needs of technology-based enterprises with the product supply of financial institutions, reducing information search and negotiation costs. In the link of operation and management, AI technology has replaced some repetitive manual operations (Ma & Han, 2025). Meanwhile, AI has broken geographical and channel constraints, extending technology finance services to county-level regions and remote areas, and effectively expanding the service coverage.

The long-standing information asymmetry in the technology finance sector can be effectively alleviated through AI technology. Big data analytics can integrate multi-dimensional fragmented data such as enterprises' R&D data and patent

information, providing a quantitative basis for financial institutions' investment decisions (Mestikou et al., 2023). Meanwhile, AI enables the optimization of technology finance risk management and control (Danielsson et al., 2022; Li & Gao, 2025): it establishes dynamic risk early warning models to track enterprises' R&D progress, capital utilization efficiency, changes in the market environment, and other factors in real time, identifying potential risks in advance and significantly reducing the credit default rate (Chen et al., 2022). Furthermore, it attracts more social capital into the technology finance sector, expands the scale of capital supply, and forms a positive cycle of capital flow.

AI equips technology finance with the capability to process and analyze massive volumes of data (Sun et al., 2019), excavating valuable information from data and enhancing the precision of capital allocation. By accurately identifying innovation potential and market demand (Agrawal et al., 2019), AI optimizes the regional and industrial allocation structure of technology finance resources, thereby improving resource allocation efficiency. In terms of regional allocation, AI technology reduces the constraints of geographical distance on resource flows. In terms of industrial allocation, AI algorithms can analyze industry technological trends and market prospects, guiding financial capital to concentrate on strategic emerging industries and avoiding excessive capital inflow into low-value-added sectors..

2.2 The impact pathways and effects of technology finance development on AI

As a specialized financial system supporting technological innovation, technology finance provides guarantees for the development of the AI industry through capital supply, risk sharing, and ecological support, and serves as the core underpinning for the high-quality development of the AI industry.

Technology finance offers multi-level capital supply for AI development. The AI industry is characterized by high investment, long cycles, and high risks, with differentiated capital demands across different development stages. technology finance has established a comprehensive capital support system through diversified financial instruments. It provides long-term and stable capital support for breakthroughs in AI's underlying technologies; for instance, sci-tech credit addresses the financing dilemmas of AI enterprises—typically asset-light and R&D-intensive—through innovative products such as credit loans and intellectual property pledges (Ye et al., 2022).

The high riskiness of the AI industry has led to widespread investment hesitations among market capital. Technology finance reduces the cost of innovation risks through approaches such as risk compensation, insurance coverage, and guarantee and credit enhancement (Zhu et al., 2024). In terms of risk compensation, it provides a certain proportion of compensation for non-performing loan losses incurred by financial institutions when extending loans to AI enterprises, incentivizing credit provision. In terms of insurance coverage, it offers comprehensive risk protection for AI enterprises' R&D activities, technological achievements, and market promotion. In terms of guarantee and credit enhancement, government-led financing guarantee institutions provide low-cost guarantee services for AI enterprises, amplifying the leverage effect of sci-tech credit and helping enterprises break through financing guarantee bottlenecks.

Through resource integration and platform construction, technology finance promotes the formation of a collaborative innovation network involving AI enterprises, universities, research institutes, and financial institutions. On one hand, technology finance institutions facilitate cooperation between AI enterprises and upstream/downstream enterprises through industrial chain investment, co-incubation, and other methods; on the other hand, platform carriers such as sci-tech and financial reform pilot zones and technology finance service centers provide integrated support for various innovation entities, including information sharing, technology matching, and financing services (Yang et al., 2025). In addition, centered on serving technological innovation and industrial upgrading, the transaction data, risk data, and demand data generated from technology finance businesses provide real-scenario support for AI algorithm optimization and model training, driving the iteration of AI technology toward greater precision, security, and efficiency.

3 Research methodology and data sources

3.1 Construction of indicator system and data sources

3.1.1 Measurement of technology finance development index

This study draws on the research of Zou et al. (2024) and categorises technology finance into public technology finance and market-oriented technology finance for measurement. The specific sub-indicators and calculation methods are presented in Table 1. Furthermore, this study adopts the entropy method to measure the technology finance development level of each province in China. As an objective weighting method, the entropy method derives weights from data dispersion, which can effectively eliminate interference from subjective factors.

Table 1. Comprehensive evaluation indicator system for technology finance development and description

Level-1 indicator	Level-2 indicator	Level-3 indicator	Calculation method	Attribute
Technology finance	Public technology finance	Fiscal Expenditure on S&T per Capita	Fiscal Expenditure on S&T / Total Population	Positive
		Intensity of Fiscal Expenditure on S&T	Fiscal Expenditure on S&T / GDP	Positive
		Government Funds for R&D per Capita	Government R&D Expenditure / Total Population	Positive
		Non-governmental Funds for R&D per Capita	Non-governmental R&D Expenditure / Total Population	Positive
	Market-oriented technology finance	Technological Upgrading Expenditure per Capita	High-tech Industry Technological Upgrading Expenditure / Total Population	Positive
		High-tech Industry Investment per Capita	High-tech Industry Investment / Total Population	Positive
		Scale of High-tech Industry	Number of High-tech Industry Enterprises	Positive
		Venture Capital Intensity	Venture Capital Investment / GDP	Positive

3.1.2 Measurement of AI development level indicators

This study draws on relevant studies and measures AI development level using the ratio of the number of AI invention patents and utility model patents applied for in a region in the current year to the total number of patents and utility model patents applied for in that region. AI patents are identified by constructing an AI lexicon and matching AI-related keywords.

3.2 Research methodology

3.2.1 Coupling coordination degree model

Considering the significant synergistic and mutually reinforcing characteristics between the development of AI and technology finance, this study introduces the coupling coordination degree model to conduct an empirical analysis of their intrinsic correlation mechanism, aiming to scientifically depict the dynamic characteristics and evolutionary laws of the coordinated development of the two systems. This model possesses unique advantages in measuring the degree of interdependence, synergistic evolution, and coordination status among multiple systems, and can accurately reflect the interactive relationships between systems. Drawing on the mature methodological framework proposed in existing literature (Shen & Jin, 2025), this study conducts a precise quantitative measurement of the coupling coordination degree between AI and technology finance development. The specific model construction and measurement process are as follows:

$$C = \frac{\sqrt[2]{X_D W_T}}{X_D + X_T} \quad (1)$$

$$S = \sqrt{C \cdot T} \quad (2)$$

$$T = \alpha X_D + \beta X_T \quad (3)$$

In the formula: X_D denotes the AI development level of a province, represented by the range-standardised value of the AI index, with a range of $[0, 1]$; X_T denotes the technology finance development level of a province, represented by the range-standardised value of technology finance development level data, with a range of $[0, 1]$. α and β are the weights assigned to the two dimensions, where $\alpha = \beta = 0.5$ and their sum equals 1. C denotes the coupling degree between AI and technology finance development; T denotes the comprehensive coordination index; S denotes the coupling coordination degree of the two systems, with a range of $[0, 1]$. The closer S is to 1, the higher the degree of synergy between AI and technology finance; the closer S is to 0, the lower the degree of synergy between the two.

3.2.2 Modified gravity model

The inter-provincial spatial correlation network of coupling coordination between AI and technology finance development in China constructed in this study takes provinces as nodes and inter-provincial spatial correlation relationships as edges; this network constitutes the set of such relationships. Drawing on the relevant research of Zhang et al. (2025), this study employs the modified gravity model to construct a spatial correlation matrix of coupling coordination between AI and technology finance development, and measures the spatial correlation intensity of coupling coordination between AI and technology finance development. The calculation formula is as follows:

$$R_{ij} = k_{ij} \frac{\sqrt[3]{G_i P_i L_i} \sqrt[3]{G_j P_j L_j}}{\left(\frac{D_{ij}}{g_i - g_j} \right)^2}, k_{ij} = \frac{L_i}{L_i + L_j} \quad (4)$$

In the formula: R_{ij} denotes the spatial correlation intensity of coupling coordination between AI and technology finance development between province i and province j ; K_{ij} represents the gravity coefficient between province i and province j ; G_i and G_j denote the GDP of province i and province j , respectively; P_i and P_j denote the permanent population of province i and province j , respectively; L_i and L_j denote the coupling coordination degree of AI and technology finance development in province i and province j , respectively; D_{ij} represents the distance between the provincial capital cities of the two provinces; g_i and g_j denote the per capita GDP of province i and province j , respectively.

3.2.3 Social network analysis method

This study draws on the research of scholars such as Yang & Su (2024) and adopts the Social Network Analysis method to analyse the spatial correlation characteristics of coupling coordination between AI and technology finance development. It achieves a holistic characterisation of relational data from three dimensions: overall, individual, and clustered network structural characteristics.

In terms of overall network structural characteristics: network density reflects the density of correlations between the coupling coordination degrees of AI and technology finance development across provinces, with a higher value indicating stronger inter-regional correlations; network connectivity reflects the robustness of the spatial correlation network itself; network hierarchy measures the asymmetric reachability of each province, with a higher value indicating that the network is dominated by a small number of provinces; network efficiency reflects the number of connections, with a lower value indicating more connections and stronger network stability.

Individual network structural characteristics are reflected by centrality indicators. Degree centrality measures the centrality of each province, with a higher value indicating a higher central position and influence of the province in the network. Closeness centrality characterises the autonomy of each province, with a higher value indicating that the province is more likely to establish connections with the outside world. Betweenness centrality reflects the intermediary role of each province in the network, with a higher value indicating a stronger ability to control and regulate other provinces.

Clustered network structural characteristics are analysed using block model analysis, which reveals node position characteristics, internal network structure, and node roles through spatial clustering—grouping nodes with identical roles into the same block. Drawing on existing research, four blocks can be identified: the net spillover block, the net benefit block, the bidirectional spillover block, and the broker block. The calculation formulas of specific indicators and the criteria for block division are presented in Table 2.

Table 2. Characteristic indicators and descriptions of the spatial association network

Indicator		Calculation formula	Description
Overall	Network	$D = \frac{L}{N(N-1)}$	L is the number of actual association relationships between provinces in the spatial network; N is the number of provinces in the spatial association network (same below). Reflects the closeness of spatial network associations for the coupling and coordination of AI and S&T finance development.
Network	Density		

(5)

Individual Network	Network Connectedness	$C = 1 - \frac{V}{N(N-1)/2} \quad (6)$	V is the number of unreachable province pairs in the relationship network. Measures the strength of connectivity between nodes in the network; a higher value indicates better network connectivity.
	Network Hierarchy	$H = 1 - \frac{K}{\max K} \quad (7)$	K is the number of reachable node provinces in the low-altitude economy network, and maxK is its maximum value. Reflects the degree of asymmetry in network relationships; a higher value indicates stronger dominance.
	Network Efficiency	$E = 1 - \frac{M}{\max M} \quad (8)$	M is the number of redundant association relationships. Indicates the quantity of excess connections in the network; lower efficiency suggests more redundant associations.
	Degree Centrality	$LD = \frac{n}{N-1} \quad (9)$	n is the total number of spatial associations. A higher degree centrality value indicates the node is more centrally positioned and has stronger control over other nodes.
	Closeness Centrality	$CC = \sum_{j=1}^N d_{ij} \quad (10)$	d_{ij} is the geodesic distance between two provinces. A province with greater closeness centrality reflects a stronger ability to avoid control by other provinces.
	Betweenness Centrality	$MC = \frac{2 \sum_j \sum_k \sum_i g_{jk}(i)}{N^2 - 3N + 2} \quad (11)$	g_{jk} is the number of geodesics between provinces j and k, $\frac{g_{jk}(i)}{g_{jk}}$ is the probability that province i lies on the geodesic between j and k, where $j \neq k \neq i$, and $j < k$. A higher betweenness centrality indicates a stronger intermediary role for that province.
	Net Spillover Block	The number of outgoing relationships far exceeds the number of incoming relationships.	
	Net Beneficiary Block	The number of incoming external relationships far exceeds the number of outgoing relationships.	
	Bidirectional Spillover Block	Exhibits significant spillover effects both internally and externally, with a relative balance between outgoing and incoming relationships, and a net effect that is not obvious.	
	Broker Block	Has numerous associations with other blocks, playing a prominent intermediary role between different blocks.	

3.2 Data sources

Considering the completeness, availability, and continuity of data, this study selects data from 30 provinces (autonomous regions and municipalities directly under the Central Government) in China as research samples, excluding data from Tibet, Hong Kong, Macao, and Taiwan regions, and sets the observation period from 2014 to 2023. Raw data are sourced from the China Statistical Yearbook, statistical yearbooks of individual provinces, the China Science and Technology Statistical Yearbook, and the CNRDS (China National Research Data Service) database, among other sources. Partial missing data are supplemented using the linear interpolation method. Geographic distances between provinces are calculated via the Distance tool in ArcGIS software, combined with the latitude and longitude coordinates of each provincial capital city.

4 Results and analysis

4.1 Analysis of spatial correlation characteristics of Coupling coordination between AI and technology finance Development

4.1.1 Evaluation of the coupling degree between AI and technology finance Development

Based on the measured results of the coupling coordination degree between AI and technology finance development, this study analyses the characteristics across four regions: Eastern, Central, Western, and Northeast China. As shown in the measurement results in Table 3, the coupling coordination degree between AI and technology finance development at the national level ranges from 0.262 to 0.330, showing an overall trend of slight fluctuations with notable volatility in certain years—for instance, rising to a phased high of 0.330 in 2019 before declining thereafter.

The coupling coordination degree in Eastern China has consistently remained the highest nationwide: it fluctuated and rose from 0.379 in 2014, reaching a peak of 0.459 in 2020, and although it declined slightly in subsequent years, it still maintained a relatively high level of 0.379 in 2023. Its average value is significantly higher than the national average, indicating a high level of synergy between AI technological innovation and technology finance resource allocation in Eastern China.

The coupling coordination degree in Central China fluctuates between 0.214 and 0.333, slightly lower than the national average overall. This reflects that while Central China has made certain progress in promoting the synergistic development of AI and technology finance, further enhancement is needed in the future.

The coupling coordination degree in Western China lies in a relatively low range of 0.137 to 0.250, significantly below the national level. This suggests that Western China still has obvious shortcomings in constructing synergistic mechanisms between AI technology implementation and technology finance service support, leaving considerable room for improvement.

Northeast China's coupling coordination degree reached a relatively high level of 0.282 in 2014 but has since shown an overall fluctuating downward trend, indicating that the synergistic effect between AI and technology finance in this region has gradually weakened.

Overall, there are significant disparities in the coupling degree of AI and technology finance across different regions: Eastern China leads the way, the Central and Western regions lag relatively behind, and Northeast China faces challenges of decline. In the future, it is necessary to strengthen policy support and regional collaboration in a targeted manner to promote overall synergistic development.

Table 3. Coupling degree measurement and evaluation results for AI and technology finance development

Region	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
National	0.267	0.306	0.269	0.262	0.267	0.330	0.305	0.237	0.239	0.273
Eastern Region	0.379	0.409	0.377	0.371	0.38	0.414	0.459	0.345	0.353	0.379
Central Region	0.217	0.307	0.230	0.246	0.214	0.333	0.239	0.230	0.234	0.267
Western Region	0.174	0.208	0.179	0.159	0.187	0.250	0.190	0.137	0.144	0.182
Northeast Region	0.282	0.254	0.255	0.252	0.228	0.253	0.281	0.207	0.166	0.202

Figure 1 illustrates the coupling degree levels of AI and technology finance development across provinces in 2014 and 2023. From 2014 to 2023, the coupling degree of AI and technology finance in China exhibited an overall trend of "steady progress with moderate growth", yet the characteristic of inter-regional imbalance remains prominent.

Beijing, Shanghai, and Guangdong have consistently remained in the first tier of coupling degrees among all provinces nationwide, maintaining a high level from 2014 to 2023—reflecting the sustained leading position of first-tier cities and economically strong provinces in the integration of technological and financial resources. Provinces/municipalities including Zhejiang, Anhui, Jiangxi, Hunan, and Chongqing have witnessed a significant increase in coupling degree, achieving rapid improvement in the synergy level between AI and technology finance.

Provinces such as Hebei, Inner Mongolia, Gansu, Ningxia, and Xinjiang have relatively lagging coupling degrees, indicating that these regions still face bottlenecks such as weak industrial foundations and insufficient financial resources in the synergistic development of AI and technology finance.

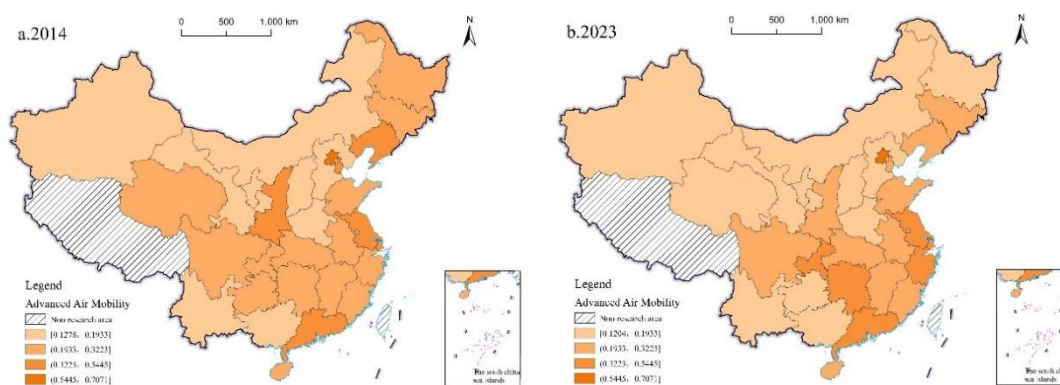


Figure 1. Spatial distribution of the coupling degree of AI and technology finance development in China

4.1.2 Analysis of overall network characteristics

To conduct an in-depth analysis of the structural characteristics of the spatial correlation network of coupling coordination between AI and technology finance development in China, this study carries out a detailed measurement and temporal analysis of the characteristics of this spatial correlation network from the perspectives of the number of network relationships, network density, network connectivity, network hierarchy, and network efficiency. Table 4 presents the evolutionary trends of the overall structural characteristics of the spatial correlation network of coupling coordination between AI and technology finance in China from 2014 to 2023.

In terms of the number of network relationships, the total number of relationships remained stable over the decade with a small fluctuation range, varying between 193 and 211. The number of relationships dropped to a relatively low level of 193 in 2019, then gradually recovered and reached a peak of 211 in 2022. This trend reflects that the cross-regional correlation between AI and technology finance has gradually strengthened after a short-term adjustment, which may be attributed to the accelerated technological diffusion, the coordinated advancement of financial support policies, and the continuous improvement of regional cooperation mechanisms—all of which have driven the recovery and growth of connection numbers in the spatial correlation network. Regarding network density, it showed an overall trend of slight fluctuations from 2014 to 2023, with density values ranging between 0.222 and 0.243, and the overall level remains low. Although the spatial linkage of the integrated development of AI and technology finance has increased, there is still considerable room for improvement in the tightness of the overall network, indicating that the substantive connections between regions in terms of technological integration and financial coordination need to be further strengthened.

Network connectivity remained consistently at 1 throughout the study period, indicating the absence of isolated nodes in the entire network—any two regions can achieve connectivity through direct or indirect paths. This result suggests that the spatial network for the synergistic development of AI and technology finance in China as a whole possesses a highly complete correlation structure and accessibility, providing a stable network foundation for the flow and optimal allocation of factors such as technology, capital, and data nationwide.

Network hierarchy exhibited a relatively obvious downward trend: a decrease in hierarchy implies the weakening of dominance relationships between nodes in the network, and the hierarchical structure tends to flatten. This indicates that the gap in status among regions in the collaborative network of AI and technology finance has gradually narrowed, the control of traditional core regions over the network has weakened, the participation of peripheral regions has increased, and the network structure has become more balanced—all of which help enhance the stability of the overall system and the sustainability of collaborative development.

Network efficiency exhibited a limited overall change but a gradual downward trend throughout the period. A decrease in network efficiency typically indicates an increase in redundant connections, richer connectivity paths, and a more complex network structure. This change suggests that the spatial correlation network for the synergistic development of AI and technology finance is gradually optimising in structure, with more diverse and stable inter-regional connections, and enhanced network resilience and resource allocation capacity.

In summary, the overall structure of the spatial correlation network of coupling coordination between AI and technology finance in China gradually optimised from 2014 to 2023. The network maintained a high level of connectivity throughout, with no isolated provinces, providing a fundamental guarantee for the flow of factors such as technology, capital, and data; although the number of network relationships and density fluctuated, the overall collaborative framework gradually took shape; the overall decline in network hierarchy reflected a moderation of status disparities between regions and a more balanced network; meanwhile, the slight fluctuations in network efficiency reflected diversified adjustments in connection methods and an improvement in network resilience. This structural evolution aligns with the national policy orientation of continuously promoting the synergistic development of AI and financial innovation and strengthening regional technology finance cooperation.

Table 4. Overall structural characteristics of the spatial correlation network of coupling coordination between AI and technology finance development

Year	Number of network relationships	Network density	Network connectedness	Network hierarchy	Network efficiency
2014	209	0.240	1	0.340	0.672
2015	204	0.235	1	0.292	0.687
2016	206	0.237	1	0.298	0.678
2017	205	0.236	1	0.297	0.675
2018	205	0.236	1	0.297	0.670
2019	193	0.222	1	0.297	0.697
2020	195	0.224	1	0.296	0.690
2021	204	0.235	1	0.296	0.675
2022	211	0.243	1	0.245	0.665
2023	207	0.238	1	0.297	0.668

Based on the Modified Gravity Model, this study measured the coupling coordination degree between the development of AI and technology finance in China from 2014 to 2023, and constructed a matrix reflecting the coordination level accordingly. By conducting binary processing on this matrix, a gravity matrix for depicting the inter-provincial spatial correlation structure was further constructed. The year 2014 and 2023 were selected as observation time points, and with the help of ArcGIS and UCINET tools, the spatial correlation intensity maps and spatial network structure maps of coupling coordination between AI and technology finance development in China for 2014 and 2023 were drawn respectively (Figures 2 and 3).

In terms of spatial correlation intensity, during the study period, the structure of the spatial correlation network for the coupling coordination between AI and technology finance has gradually become more complex and dense over time, with spatial correlation continuing to strengthen. In 2014, the overall pattern was characterised as "dense in the East and loose in the Central and Western regions". The correlation lines among eastern coastal provinces were thick and densely distributed, reflecting strong potential for coupling coordination links between provinces in these regions in terms of AI and technology finance development; in contrast, the correlation lines in the Central, Western, and Northeast China were mostly thin and scattered, and except for a few core provincial capital cities, the coupling coordination links between most provinces and other regions were generally weak.

By 2023, the density of the spatial correlation network had increased significantly. The correlation lines in the eastern region became even denser, and in some areas, the correlation lines thickened and became more concentrated. Previously sparse regions gradually formed local correlation clusters, and both the coverage and tightness of the spatial correlation between AI and technology finance development improved markedly.

In terms of spatial network structure, provinces including Beijing, Shanghai, Jiangsu, and Zhejiang have become core nodes with the densest network connections, undertaking the functions of resource integration and linkage radiation in the coupling coordination between AI and technology finance, and constituting the dynamic hub of the spatial correlation network. Compared with 2014, provinces such as Fujian, Hubei, and Guangdong saw a significant improvement in network tightness and centrality indicators in 2023, with their node status rising continuously—they have developed into secondary core nodes with strong radiation capacity. Meanwhile, the radiation scope of core nodes has expanded continuously toward the Central and Western regions, and connections with these regions have become increasingly close. This indicates that the participation and importance of the Central and Western regions in the national synergistic pattern of AI and technology finance development are constantly rising.

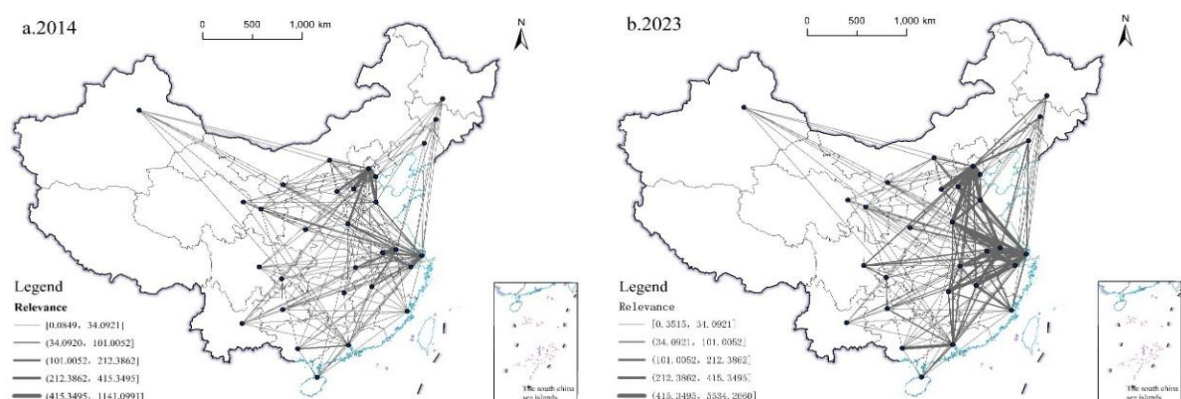


Figure 2. Spatial association intensity of the coupling coordination between AI and technology finance development

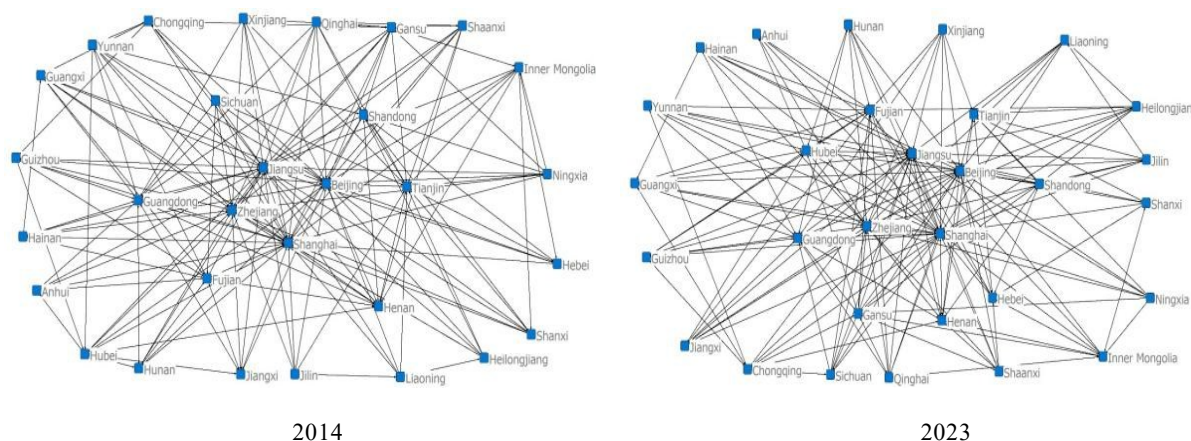


Figure 3. Spatial network of the coupling coordination between AI and technology finance development in China

4.1.3 Analysis of individual network structure

This study conducts an in-depth analysis of the individual structural characteristics of the spatial correlation network of coupling coordination between AI and technology finance development from three dimensions: degree centrality, betweenness centrality, and closeness centrality, to reveal the functional roles and importance of different provinces in the network.

As shown in Table 5, in terms of degree centrality, Shanghai, Beijing, and Jiangsu have consistently maintained leading positions, with values significantly higher than those of other provinces. This indicates that these municipalities and provinces have long occupied a core hub position in the collaborative network of AI and technology finance, with extensive direct connections and outstanding capacity for the agglomeration and radiation of resource factors. Some provinces have exhibited a significant upward trend. For instance, Hebei's degree centrality increased from 20.690 to 31.034, and Fujian's rose markedly from 37.931 to 55.172, reflecting the rapid expansion of the breadth of direct correlations and influence of these provinces in the network. In contrast, Tianjin's degree centrality decreased significantly, falling from 51.724 to 34.483, which may be affected by changes in the regional competitive landscape or adjustments to its own industrial structure. Overall, the hierarchical structure of the network remains evident: most Central and Western provinces, such as Shanxi, Jiangxi, and Hainan, have maintained low levels of degree centrality over the years, indicating that they have established limited direct connections in the network and occupy a relatively peripheral position.

The betweenness centrality indicator reveals the "bridge" role of each province in controlling resource flows within the network. In 2014, the betweenness centrality of Beijing, Shanghai, Jiangsu, Zhejiang, and Guangdong was far higher than the national average, with these regions occupying key intermediary positions. By 2023, although the betweenness centrality of core nodes such as Beijing, Shanghai, and Jiangsu still ranked among the top, most showed a downward trend—indicating a weakening of their absolute control over resource flows. Meanwhile, provinces including Fujian, Hubei, Henan, and Gansu witnessed a significant increase in betweenness centrality, reflecting the growing strength of their bridge role in connecting different regional blocks; the intermediary function of the network has exhibited a diversified trend of diffusion from core nodes to emerging nodes. In contrast, provinces such as Anhui and Jiangxi have consistently maintained low levels of betweenness centrality, acting more as factor "recipients" rather than "coordinators" in the network.

Closeness centrality reflects a node's ability to obtain resources independently without being controlled by others. Provinces including Shanghai, Beijing, Jiangsu, and Zhejiang have consistently led other provinces in closeness centrality, indicating that they are located in the core circle of the network, can rapidly establish connections with other nodes in the network, and possess extremely high autonomy and independence. Notably, provinces such as Hebei, Henan, Hubei, and Fujian have seen a marked increase in closeness centrality, which suggests a reduction in their dependence on external nodes and enhanced autonomy and structural status. Provinces including Anhui, Shanxi, Jiangxi, and Xinjiang have consistently maintained low levels of closeness centrality, meaning they remain in relatively peripheral and passive positions in the network—they

need to rely on longer paths or more intermediaries to obtain resources and are vulnerable to fluctuations in the network structure.

Table 5. Individual structural characteristics of the spatial association network for the coupling coordination between AI and technology finance

Province	Degree Centrality		Betweenness Centrality		Closeness Centrality	
	2014	2023	2014	2023	2014	2023
Beijing	79.310	82.759	12.522	13.816	82.857	85.294
Tianjin	51.724	34.483	3.395	1.401	65.909	59.184
Hebei	20.690	31.034	0.095	0.844	55.769	59.184
Shanxi	24.138	20.690	0.208	0.142	56.863	55.769
Inner Mongolia	31.034	31.034	0.400	0.547	59.184	59.184
Liaoning	20.690	24.138	0.109	0.192	55.769	56.863
Jilin	20.690	24.138	0.174	0.238	55.769	56.863
Heilongjiang	24.138	27.586	0.251	0.530	56.863	58.000
Shanghai	93.103	93.103	15.266	14.249	93.548	93.548
Jiangsu	93.103	89.655	15.266	13.386	93.548	90.625
Zhejiang	68.966	68.966	6.956	6.676	76.316	76.316
Anhui	20.690	17.241	0.124	0.032	54.717	53.704
Fujian	37.931	55.172	1.369	4.117	61.702	69.048
Jiangxi	24.138	24.138	0.214	0.149	56.863	56.863
Shandong	41.379	31.034	1.251	0.555	63.043	59.184
Henan	37.931	41.379	1.311	1.724	61.702	63.043
Hubei	34.483	44.828	0.842	1.379	60.417	64.444
Hunan	24.138	24.138	0.214	0.149	56.863	56.863
Guangdong	48.276	48.276	2.669	1.939	65.909	65.909
Guangxi	27.586	31.034	0.325	0.376	58.000	59.184
Hainan	24.138	24.138	0.152	0.125	56.863	56.863
Chongqing	27.586	31.034	0.382	0.462	58.000	59.184
Sichuan	27.586	27.586	0.604	0.218	58.000	58.000
Guizhou	27.586	27.586	0.375	0.274	58.000	58.000
Yunnan	31.034	27.586	0.603	0.274	59.184	58.000
Shaanxi	27.586	24.138	0.171	0.206	58.000	56.863
Gansu	44.828	48.276	1.464	2.012	64.444	65.909
Qinghai	31.034	27.586	0.360	0.256	59.184	58.000
Ningxia	27.586	24.138	0.208	0.329	58.000	56.863
Xinjiang	24.138	24.138	0.207	0.397	56.863	56.863
Mean	37.241	37.701	2.250	2.233	62.605	62.787

4.1.4 Analysis of Block Model and Block Clustering Characteristics

To conduct an in-depth analysis of the spatial correlation network structure for the synergistic development of AI and technology finance, this study draws on existing research methods and employs the CONCOR module in UCINET software to divide the correlation network into four functional blocks: Block I (Net Benefit Block), Block II (Broker Block), Block III (Bidirectional Spillover Block), and Block IV (Net Spillover Block). Table 6 presents the correlation relationships of each block in 2014 and 2023, revealing the dynamic evolution of the network structure.

In 2014, the spatial correlation network for the synergistic development of AI and technology finance was dominated by inter-block spillovers. Block I's actual internal relationship ratio of 61.11% was far higher than the expected value of 13.79%, and its 92 external-block relationships received were significantly more than the 18 external-block relationships sent, demonstrating a typical "net benefit" attribute. Block II's actual internal relationship ratio of 3.23% was significantly lower than the expected value of 10.34%, and its received and sent relationships were mainly directed toward external blocks,

fulfilling the “broker” role of connecting different blocks. Block III’s actual internal relationship ratio of 14.47% was lower than the expected value of 37.93%, and its 70 external-block relationships sent were far more than the 10 external-block relationships received, acting as the “bidirectional spillover block”. Block IV had an extremely low actual internal relationship ratio of 1.75%, and its 60 external-block relationships sent were significantly more than the 37 received from external blocks, making it the main “net spillover block” in the network.

By 2023, the network block structure had undergone significant evolution. Block I’s “net benefit” attribute remained stable: the number of external-block relationships it received increased to 101, and its actual internal relationship ratio of 30.00% was still higher than the expected value of 17.24%, indicating that its reliance on external network resources has continued to deepen. Block III’s actual internal relationship ratio increased from 14.47% to 18.64%, narrowing the gap with the expected value; meanwhile, the number of external-block relationships it sent decreased to 48, and the relationships received from external blocks increased to 21. In contrast, Block IV’s net spillover effect became more prominent: its actual internal relationship ratio further decreased to 4.82%, far lower than the expected value of 34.48%, and external spillover relationships surged to 79, consolidating its position as the network’s primary “net spillover block”.

Table 6. Spillover effects of spatial association blocks for the coupling coordination between AI and technology finance

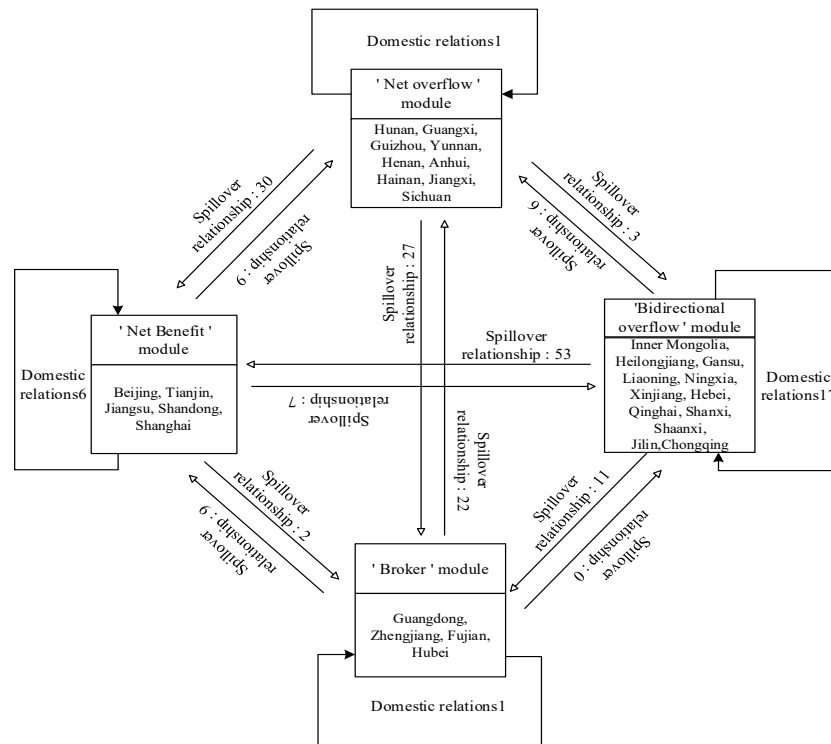
Year	Block	Received Relationships		Sent Relationships		Expected Intra-block Proportion (%)	Actual Intra-block Proportion (%)	Block Attribute
		Intra-block	Extra-block	Intra-block	Extra-block			
2014	BlockI	6	92	6	18	13.79	61.11	Net Benefit Block
	BlockII	1	40	1	31	10.34	3.23	Broker Block
	BlockIII	17	10	17	70	37.93	14.47	Two-way Spillover Block
	BlockIV	1	37	1	60	27.59	1.75	Net Spillover Block
2023	BlockI	9	101	9	21	17.24	30.00	Net Benefit Block
	BlockII	1	34	1	34	10.34	2.86	Broker Block
	BlockIII	11	21	11	48	27.59	18.64	Two-way Spillover Block
	BlockIV	4	26	4	79	34.48	4.82	Net Spillover Block

Figure 4 presents a visual analysis of the correlation relationships among spatial correlation blocks of coupling coordination between AI and technology finance development. In 2014, the Net Benefit Block included Beijing, Tianjin, Jiangsu, Shandong, and Shanghai; by 2023, Inner Mongolia and Zhejiang were added to the Net Benefit Block, indicating an enhancement in their resource absorption capacity. Most regions in the Net Benefit Block boast strong economic strength, a high agglomeration of scientific and technological innovation resources, and well-developed AI R&D systems and technology finance service networks. They not only maintain close internal connections but also absorb substantial external resources through spillover relationships—this attractiveness is closely linked to their outstanding technology transfer capacity, sufficient capital supply capacity, and abundant AI industrial application scenarios.

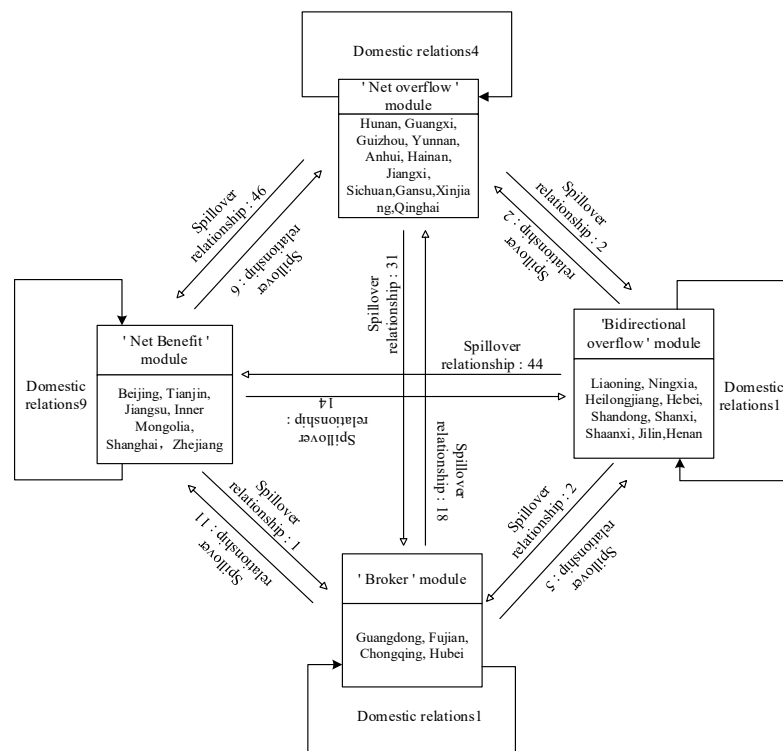
In 2014, the Broker Block covered Guangdong, Zhejiang, Fujian, and Hubei; in 2023, the Broker Block retained Guangdong, Fujian, and Hubei, with Chongqing added as a new member. The inclusion of Chongqing reflects the further strengthening of its intermediary function: Chongqing can link core eastern STI and financial resources with western industrial scenarios, improving cross-block resource allocation efficiency. Provinces in the Broker Block have a foundation for linking scientific and technological innovation resources and financial services. Guangdong, relying on the Pearl River Delta high-tech industrial clusters, and Hubei, leveraging its advantage in scientific and educational resources, are able to perform the “intermediary” function between different blocks, facilitating the cross-regional flow and matching of resources.

In 2014, the Bidirectional Spillover Block included provinces such as Ningxia, Inner Mongolia, Liaoning, and Hebei; provinces within the block maintained frequent connections with each other and simultaneously sustained bidirectional resource flow with other blocks. In 2023, the provinces in the Bidirectional Spillover Block underwent significant adjustments. On the one hand, regions with solid industrial foundations (e.g., Liaoning and Hebei) can export intelligent transformation technologies and industrial supporting resources; on the other hand, characteristic industrial scenarios in areas such as Ningxia provide implementation carriers for AI and technology finance, enabling bidirectional resource interaction.

In 2014, the Net Spillover Block covered Central and Western provinces including Guangxi, Yunnan, and Sichuan. By 2023, the number of provinces in the Net Spillover Block had increased. Scenarios such as agricultural digitalisation and characteristic manufacturing in Central and Western provinces continuously serve as implementation carriers for AI and technology finance nationwide, maintaining a long-term position as resource exporters.



2014



2023

Figure 4. Interrelationships among the four blocks in the spatial network of the coupling coordination between AI and technology finance development

Table 7 presents the density matrices and image matrices of the spatial correlation network of coupling coordination between AI and technology finance for 2014 and 2023. The density matrix reflects the actual correlation intensity between blocks, while the image matrix reveals significant spillover paths and structural interdependence relationships.

In terms of the density matrix, intra-block correlation exhibited unbalanced characteristics in 2014. The intra-block density of Block I (0.550) and Block III (0.129) were both higher than those of other blocks, indicating relatively close internal connections. Among inter-block correlations, the density of Block III to Block I reached the highest value of 0.883, demonstrating extremely strong directional spillover intensity; meanwhile, the density of Block IV to Block II also reached 0.750, indicating close interaction between the two. By 2023, the inter-block correlation pattern had undergone significant evolution. The intra-block density of Block I decreased to 0.300, while that of Block II remained at a low level of 0.083, showing weak internal connections for both blocks. Although the density of Block III to Block I decreased slightly, it still maintained a high level of 0.815, continuing to serve as an important external influence source for Block I; the correlation intensity of Block IV to Block II remained prominent, and the density of Block IV to Block I increased.

The image matrix further reveals the significant spillover transmission structure between blocks. In 2014, spillover paths between blocks were relatively concentrated. Block I only received spillovers from Block III and Block IV, without generating significant outward spillovers itself. Block IV was the most active spillover source in the network, exerting significant spillovers on both Block I and Block II. Block II played a unique intermediary role: it received spillovers from Block IV but did not form direct spillovers to other blocks. By 2023, the network spillover structure exhibited more complex multi-path characteristics. Block I's role underwent a transformation, evolving from a pure beneficiary to a node that emits spillovers to both Block II and Block III; however, its primary role remained receiving external spillovers, particularly multi-directional inflows from Block II, III, and IV. Block II's intermediary hub position was strengthened: it not only received spillovers from Block IV but also emitted spillovers to both Block I and Block IV, acting as a key “bridge” between multiple blocks.

Table 7. Density matrices and image matrices of the spatial correlation network of coupling coordination between AI and technology finance Development

Year	Block	Density matrices				Image matrices			
		BlockI	BlockII	BlockIII	BlockIV	BlockI	BlockII	BlockIII	BlockIV
2014	BlockI	0.550	0.100	0.117	0.200	1	0	0	0
	BlockII	0.450	0.083	0.000	0.611	0	0	0	1
	BlockIII	0.883	0.229	0.129	0.056	1	0	0	0
	BlockIV	0.667	0.750	0.028	0.014	1	1	0	0
2023	BlockI	0.300	0.042	0.259	0.091	1	0	1	0
	BlockII	0.458	0.083	0.139	0.409	1	0	0	1
	BlockIII	0.815	0.056	0.153	0.020	1	0	0	0
	BlockIV	0.697	0.705	0.020	0.036	1	1	0	0

4.2 Analysis of driving factors of the spatial correlation network of coupling coordination between AI and technology finance development

4.2.1 Model construction

This study adopts the QAP regression model to explore the driving factors after analysing the structural characteristics of the spatial correlation network. The spatial correlation network of coupling coordination between AI and technology finance development is composed of relational data matrices, which suffers from multicollinearity and autocorrelation issues—making it impossible to analyse correlations using traditional statistical test methods. To address this, this study refers to relevant research (Li et al., 2024) and employs the QAP regression model for empirical testing to further analyse the driving factors of the spatial correlation network of coupling coordination between AI and technology finance development. This method avoids

traditional econometric constraints through the non-parametric characteristics of matrix permutation. The model is more robust than traditional parametric tests and can avoid biases caused by multicollinearity.

Drawing on existing studies and the characteristics of AI and technology finance development, the driving factors are selected as follows: (1) Geographic spatial proximity (Dis): An inter-provincial adjacency weight matrix is used to characterise geographic spatial adjacency, where 1 is assigned if provinces are adjacent and 0 if not. (2) Difference in innovation level (Inn): It is measured by the difference in the number of domestic patent applications accepted. (3) Difference in the development level of the technology market (Tec): It is characterised by the difference in the ratio of technology market turnover to regional gross domestic product (GDP). (4) Difference in economic development level (ED): It is measured by the difference in per capita GDP. (5) Difference in urbanization level (Urb): It is characterised by the difference in the proportion of urban population to total population.

Based on this, the QAP regression model is constructed as follows:

$$AT = F(\text{Dis}, \text{Inn}, \text{Tec}, \text{ED}, \text{Urb}) \quad (12)$$

In the formula: AT denotes the gravity matrix of the spatial correlation network for the coupling coordination between AI and technology finance development; Dis represents the spatial adjacency weight matrix; Inn, Tec, ED, and Urb respectively denote the difference matrices constructed after standardisation of each driving factor.

4.2.2 QAP correlation analysis

Based on the 2014 data, this study examines the formation mechanism of the spatial correlation network for the coupling coordination between AI and technology finance development. Table 8 presents the results of the QAP correlation analysis: the correlation coefficients between the spatial correlation matrix and the difference matrix of innovation level, difference matrix of technology market development level, difference matrix of economic development level, geographic spatial adjacency matrix, and difference matrix of urbanisation level are 0.291, 0.245, 0.466, 0.179, and 0.386 respectively. All coefficients have passed the statistical significance test, indicating that the above factors exert significant impacts on the formation process of the spatial correlation network.

Table 8. QAP correlation results between the spatial correlation matrix of coupling coordination development of AI and technology finance and driving factors

Variables	Correlation coefficient	Mean correlation coefficient	Standard value	Minimum	Maximum	$P_1 \geq 0$	$P_2 \leq 0$
Inn	0.291***	-0.002	0.068	-0.148	0.272	0.000	1.000
Tec	0.245***	0.001	0.078	-0.107	0.273	0.002	0.999
ED	0.466***	0.001	0.067	-0.141	0.289	0.000	1.000
Dis	0.179***	0.000	0.038	-0.130	0.164	0.000	1.000
Urb	0.386***	0.000	0.065	-0.162	0.264	0.000	1.000

To accurately identify the driving mechanism underlying the formation of the spatial correlation network for the coupling coordination between AI and technology finance development, Table 9 presents the results of the correlation test between the difference matrices of key influencing factors. The analysis shows that there is a generally significant correlation between the difference matrices corresponding to each explanatory variable, which reveals a potential multicollinearity issue. If traditional econometric regression methods are used, multicollinearity will lead to distorted results of statistical significance tests, making it difficult to accurately assess the true impact of each factor.

Therefore, when exploring the formation mechanism of the spatial correlation network for the coupling coordination between AI and technology finance development, this study adopts the QAP regression method to mitigate issues such as estimation bias, ensuring the robustness and credibility of the research conclusions.

Table 9. QAP correlation results among spatial correlation driving factors of coupling coordination between AI and technology finance development

Variables	Inn	Tec	ED	Dis	Urb
Inn	1.000***	0.333**	0.407**	-0.075*	0.161
Tec	0.333**	1.000***	0.681***	-0.072**	0.540***

ED	0.407**	0.681***	1.000***	-0.148***	0.836***
Dis	-0.075*	-0.072**	-0.148***	1.000***	-0.158***
Urb	0.161	0.540***	0.836***	-0.158***	1.000***

4.2.3 QAP regression analysis

This study conducts an empirical analysis of the driving factors of the spatial correlation network of coupling coordination between AI and technology finance development based on the QAP regression method.

The number of random permutations is set to 20000, and the regression results are presented in Table 10, with detailed analysis as follows:

Technological Innovation Level (Inn): It exhibited a significant positive driving effect in both 2014 and 2023, with standardized coefficients of 0.156 and 0.141 respectively—both significant at the 1% level. This indicates that the gap in technological innovation levels between regions can effectively promote the formation of spatial network correlations. Regions with active technological innovation are more likely to generate knowledge spillovers and technology diffusion; by sharing R&D achievements, co-constructing innovation platforms, and other means, they strengthen inter-regional technological linkages and industrial synergy, providing core motivation for the formation of the spatial correlation network.

Difference in Technology Market Development Level (Tec): It showed a significant negative impact in both years, with standardized coefficients of -0.162 and -0.171 respectively. This implies that an expansion of the gap in technology development levels inhibits inter-provincial spatial correlations. When the gap in technology development levels between regions is excessively large, the cross-regional adaptability of AI technology decreases, and the support of technology finance for technology implementation struggles to form synergy—leading to higher barriers to cooperation and docking, which in turn hinders the cross-regional flow of factors and the tight construction of the correlation network.

Difference in Economic Development Level (ED): It is a key facilitating factor driving spatial correlations, with standardised coefficients of 0.475 and 0.470 in 2014 and 2023 respectively—both highly significant at the 1% level. This indicates that a moderate gradient in economic development helps form a pattern of complementary development between regions. Advantages of economically developed regions in technology, capital, and talent can radiate to developing regions, generating effective spillover effects and promoting the optimal allocation and synergistic linkage of AI and technology finance factors on a broader scale.

Geographic Spatial Proximity (Dis): Its impact on the spatial correlation network was significantly positive in both 2014 and 2023, with standardised coefficients of 0.264 and 0.270 respectively, both significant at the 1% level. This suggests that geographically adjacent provinces are more likely to form spatial correlation networks. Geographic proximity reduces communication costs between regions, facilitates the flow of innovative factors such as technology and talent, and provides convenient conditions for cross-regional cooperation. Additionally, adjacent regions are more likely to generate synergistic effects in policy coordination and industrial supporting facilities, thereby strengthening the construction of spatial correlation networks.

Urbanisation Level (Urb): The impact of differences in urbanisation level on the spatial correlation network shows a declining trend. In 2014, its coefficient was 0.093, significant at the 5% level; however, by 2023, the coefficient dropped to 0.051 and failed to pass the significance test. The reason may lie in that the complementarity of consumer markets and industrial layout brought by differences in urbanisation level once promoted inter-provincial linkage between AI and technology finance. But with the overall improvement and layout optimisation of national urbanisation level, the complementary effect of such differences has gradually weakened, and its driving effect on spatial correlations is no longer significant.

In summary, geographic spatial proximity, differences in innovation level, and differences in economic development level positively affect the construction of the spatial correlation network of coupling coordination between AI and technology finance development, while differences in technology market development level show a negative impact. Although differences in urbanisation level positively affect the formation of the spatial correlation network, their influence has gradually weakened. Currently, efforts should be made to build a more balanced financial ecosystem, strengthen multi-polar technology diffusion, and while respecting regional gradient differences, promote the formation of a new pattern of cross-regional development featuring complementary advantages and synergistic linkage.

Table 10. QAP regression results of driving factors for the spatial association of coupling coordination between AI and technology finance development

Driving factors	2014	2023
	Standardized coefficient	Standardized coefficient
Inn	0.156***	0.141***
Tec	-0.162***	-0.171***
ED	0.475***	0.470***
Dis	0.264***	0.270***
Urb	0.093**	0.051
R2	0.311	0.297
Adj R2	0.308	0.294
P-value	0.000***	0.000***
Observations	870	870
Random Permutations	20000	20000

5 Conclusions and recommendations

5.1 Research conclusions

The coupling coordination between AI and technology finance is a core pathway to activate innovation dynamics and empower the high-quality development of the real economy in the era of the digital economy. Clarifying the characteristics of the spatial correlation network of their coupling coordination helps promote the cross-regional optimal allocation of technology and financial resources and achieve industrial synergistic upgrading.

Based on this, this study measures the level of coupling coordination between AI and technology finance at the provincial level in China from 2014 to 2023, analyses the structural characteristics and evolutionary process of its spatial correlation network using the modified gravity model and social network analysis method, and identifies the driving factors of the correlation network with the help of the QAP regression model. The main research conclusions are as follows:

(1) The level of coupling coordination between AI and technology finance in China shows a fluctuating upward trend, with significant and dynamically evolving regional gradient differences. In terms of spatial correlation intensity, the overall pattern is characterised by “dense in the east and loose in the central and western regions”, with eastern provinces such as Shanghai and Beijing remaining at a high level of coupling coordination for a long time. Over time, the density and complexity of the spatial correlation network have gradually increased, correlation relationships have become closer, a multi-core linkage pattern has initially emerged, and the degree of inter-regional synergy has continued to deepen.

(2) The spatial correlation network of coupling coordination between AI and technology finance exhibits stable complex structural characteristics. The network has always maintained a high degree of connectivity, with no isolated provinces; the number of network relationships shows a trend of “fluctuating decline followed by recovery”, the network density has undergone minor adjustments, the network hierarchy has generally decreased, and network efficiency has fluctuated slightly. This indicates an increase in the network’s flattening degree, more diversified connection paths, a continuous enhancement of overall stability and synergy, and the gradual optimisation of the network structure.

(3) From the perspective of individual network structural characteristics, Shanghai, Beijing, Jiangsu, and other provinces have long occupied core positions in the network, leading in degree, betweenness, and closeness centrality, and undertaking the functions of resource integration and radiation-driven development; the core positions of provinces such as Fujian and Hebei have gradually improved, becoming secondary cores with strong linkage capabilities; most central and western, as well as northeastern provinces, remain in peripheral positions in the network. Although the structural gap between core and peripheral regions has eased, no fundamental change has occurred. The overall network shows a flattening development trend: the dominance relationships between nodes have weakened, and the paths of factor flow have become more complex and diversified.

(4) The spatial network of coupling coordination between AI and technology finance development can be divided into four major blocks: Net Benefit, Broker, Bidirectional Spillover, and Net Spillover. Inter-block correlations dominate among blocks with clear spillover effects. The Net Benefit Block is mainly composed of core provinces such as Beijing, Tianjin, Jiangsu, and Shanghai, continuously receiving external correlation resources; the Broker Block is dominated by provinces such as Guangdong, Fujian, and Hubei, long undertaking the intermediary function of connecting different blocks; the Bidirectional Spillover Block and Net Spillover Block are mainly central and western provinces—the external spillover effect of the Net Spillover Block has continued to strengthen with network evolution, while the spillover effect of the Bidirectional Spillover Block has declined.

(5) The formation and evolution of the spatial correlation network of coupling coordination between AI and technology finance are jointly driven by multiple factors. Differences in economic development level, geographic spatial proximity, and differences in innovation level exert a significant positive impact on the correlation network, serving as the core driving factors for the network's formation and expansion; differences in the development level of the technology market exert a stable negative impact, with the expansion of gaps in technology market maturity between regions inhibiting inter-provincial correlation intensity; the driving effect of differences in urbanisation level has gradually attenuated.

5.2 Recommendations

Based on the aforementioned research conclusions on the spatiotemporal evolution characteristics and driving factors of the coupling coordination development between AI and technology finance, the following policy recommendations are proposed to promote their in-depth integration and regional synergy:

Implement a regionally differentiated strategy and build a new pattern of gradient linkage development

Strengthen the hub radiation role of core provinces. Support core provinces such as Shanghai and Beijing in building national-level “AI + technology finance” sci-tech innovation hubs; based on sci-tech innovation finance reform pilot zones, establish cross-regional sci-tech innovation finance special funds to targetedly support AI scenario implementation projects in the central and western regions and Northeast China, and promote the gradient flow of technology and capital resources from core regions to peripheral regions.

Cultivate the linkage capacity of secondary cores: For secondary core provinces such as Fujian and Hebei, deploy regional-level “AI + Industrial Finance” linkage nodes, support the construction of provincial-level sci-tech finance service centers to undertake technology spillovers from core regions, and link with surrounding peripheral provinces to form regional cooperation circles.

Promote the integration of characteristic scenarios in peripheral regions: For peripheral provinces in the central and western regions and Northeast China, implement the “Characteristic Scenarios + Precision Finance” policy; increase the tilt of central financial transfer payments and special bonds, and focus on filling the gaps in information infrastructure and digital literacy education.

Optimise the structure of the spatial correlation network and enhance cross-regional collaborative resilience

Establish a nationwide unified docking platform for “AI + technology finance”: Integrate inter-provincial AI technology achievements and technology finance product resources to achieve intelligent matching of cross-regional technology demand and financial supply, and reduce collaboration costs across blocks and provinces. Promote functional complementarity of cross-regional network nodes: Encourage core provinces to export technology and financial resources, and peripheral provinces to provide application scenarios and carrier space.

Improve the cross-regional adaptability of digital infrastructure: Promote the integration of the “East Data West Computing” project with the digitalisation of technology finance; while laying out AI computing power sub-centers in the central and western regions, support the construction of regional technology finance data sharing nodes to achieve collaborative scheduling of cross-regional computing power, data, and financial resources, and weaken the constraints of geographic distance on the correlation network.

Focus on key driving factors and optimise policy supply and resource allocation

Establish a multi-level support system for scientific and technological innovation: Given the core driving role of

technological innovation capabilities, policy incentives for enterprises' R&D activities should be further strengthened; in particular, for small and medium-sized tech enterprises, the proportion of additional deductions for R&D expenses may be appropriately increased, and an ex-post subsidy mechanism for R&D investment should be explored. Establish national-level major special projects for AI and technology finance, and encourage the collaborative tackling of key generic technologies by the “industry-university-research-application-finance” ecosystem.

Deepen the supply-side structural reform of finance. Encourage cooperation between financial institutions and tech enterprises to develop technology finance products that serve the entire life cycle of the AI industry. On the premise of controllable risks, deepen the application of regulatory tools for technology finance innovation and promote positive interaction between innovation and regulation.

References

- Agrawal, A., Gans, J. S., & Goldfarb, A. (2019). Artificial intelligence: The ambiguous labor market impact of automating prediction. *Journal of Economic Perspectives*, 33(2), 31–50. <https://doi.org/10.1257/jep.33.2.31>
- Babina, T., Fedyk, A., He, A., & Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics*, 151, Article 103745. <https://doi.org/10.1016/j.jfineco.2023.103745>
- Bonaparte, Y. (2024). Artificial intelligence in finance: Valuations and opportunities. *Finance Research Letters*, 60, Article 104851. <https://doi.org/10.1016/j.frl.2023.104851>
- Chen, H., & Yoon, S. S. (2022). Does technology innovation in finance alleviate financing constraints and reduce debt-financing costs? Evidence from China. *Asia Pacific Business Review*, 28(4), 467–492. <https://doi.org/10.1080/13602381.2021.1874665>
- Czarnitzki, D., Fernández, G. P., & Rammer, C. (2023). Artificial intelligence and firm-level productivity. *Journal of Economic Behavior & Organization*, 211, 188–205. <https://doi.org/10.1016/j.jebo.2023.05.008>
- Danielsson, J., Macrae, R., & Uthemann, A. (2022). Artificial intelligence and systemic risk. *Journal of Banking & Finance*, 140, Article 106290. <https://doi.org/10.1016/j.jbankfin.2021.106290>
- Fan, Y. X., Meng, Y. J., & Fan, X. H. (2025). Facing the 15th Five-Year Plan: The theoretical logic, practical challenges, and implementation directions of empowering the high-quality development of the private economy with artificial intelligence. *On Economic Problems*, (11), 1–12. <https://doi.org/10.16011/j.cnki.jjw.2025.11.001>
- Guo, J., Yang, Q., & Wu, H. M. (2024). Sci-tech finance improves agricultural production efficiency: Empirical evidence from the pilot policies of sci-tech finance. *Chinese Rural Economy*, (5), 81–105.
- Hsu, P.-H., Tian, X., & Xu, Y. (2014). Financial development and innovation: Cross-country evidence. *Journal of Financial Economics*, 112(1), 116–135. <https://doi.org/10.1016/j.jfineco.2013.12.002>
- Jiang, W. X. (2025). Artificial intelligence, new quality productivity, and high-quality development of finance. *Statistics & Decision*, 41(17), 11–16. <https://doi.org/10.13546/j.cnki.tjyc.2025.17.002>
- Li, M. G., & Gao, L. A. (2025). Artificial intelligence-driven financial transformation: Opportunities, risks, and responses. *Journal of Hebei University (Philosophy and Social Science)*, 50(6), 138–146. <https://doi.org/10.3969/j.issn.1005-6378.2025.06.014>
- Li, T., Ye, A., Zhang, Y., & Zhu, L. (2024). The spatial correlation network structure and its formation mechanism of urban high-quality economic development: A comparative analysis from the Yangtze River Economic Belt and the Yellow River Basin in China. *Environmental Science and Pollution Research*, 31(16), 23728–23746. <https://doi.org/10.1007/s11356-024-32657-8>
- Lin, C. P., Wang, S. Z., Zhu, X. Y., et al. (2025). “Enabling” vs. “negating”: Impact of artificial intelligence applications on breakthrough innovation in high-tech enterprises. *R&D Management*, 1–14. <https://doi.org/10.13581/j.cnki.rdm.20241245>
- Liu, L. J., & Liu, Y. C. (2022). Study on the coupling and coordination of science and technology finance and high-quality

development of real economy in Beijing, Tianjin and Hebei. *On Economic Problems*, 44(8), 96–102. <https://doi.org/10.16011/j.cnki.jjw.2022.08.013>

Lui, A. K. H., Lee, M. C. M., & Ngai, E. W. T. (2022). Impact of artificial intelligence investment on firm value. *Annals of Operations Research*, 308(1–2), 373–388. <https://doi.org/10.1007/s10479-020-03862-8>

Ma, L. Y., & Li, X. M. (2019). Does science and technology finance policies promote regional innovation? Quasi-natural experiment based on the pilot policy of combining science and technology with finance. *China Soft Science*, (12), 30–42. <https://doi.org/10.3969/j.issn.1002-9753.2019.12.003>

Ma, T., & Han, T. C. (2025). Ethical considerations of artificial intelligence in finance: Risks and countermeasures. *Scientific Management Research*, 43(2), 156–166.

Mestikou, M. A., Smeti, K. E., & Hachaïchi, Y. (2023). Artificial intelligence and machine learning in financial services: Market developments and financial stability implications. *Financial Stability Board*.

Mishra, S., Ewing, M. T., & Cooper, H. B. (2022). Artificial intelligence focus and firm performance. *Journal of the Academy of Marketing Science*, 50(6), 1176–1197. <https://doi.org/10.1007/s11747-022-00876-5>

Pan, J. F., Wang, C. Y., & Wu, J. (2026). The essence of AI-enabled empowerment: A tripartite integration paradigm of data, knowledge, and systems. *Science Research Management*, 47(1), 1–10. <https://doi.org/10.19571/j.cnki.1000-2995.2026.01.001>

She, K. W., Shen, Y., & Zhao, S. Y. (2025). Research on credit support for technology finance: Evidence from banks' loan-level data. *Journal of Management World*, 41(9), 25–48. <https://doi.org/10.19744/j.cnki.11-1235/f.2025.0119>

Shen, J. H., & Jin, C. (2025). Coupling coordination between green finance and tourism high-quality development and its influencing factors in China. *Economic Geography*, 45(9), 239–248. <https://doi.org/10.15957/j.cnki.jjdl.2025.09.024>

Sheng, T. X., Ma, Z. P., Fan, C. L., & Teng, J. H. (2025). A study on the “leveraging effect” of science and technology finance on regional capital inflows. *Studies of International Finance*, (8), 49–62. <https://doi.org/10.16475/j.cnki.1006-1029.2025.08.002>

Sun, Y., Shi, Y., & Zhang, Z. (2019). Finance big data: Management, analysis, and applications. *International Journal of Electronic Commerce*, 23(1), 9–11. <https://doi.org/10.1080/10864415.2018.1512270>

Wang, J. X., & Zhao, M. (2025). How can “AI+” empower the high-quality development of securities firms? An explanation based on IPO underpricing. *Journal of Management World*, 41(11), 196–220.

Wang, X. H., & Liu, J. H. (2025). Sci-tech finance and total factor productivity of enterprises: Evidence from the enterprise innovation points system. *Economic Research Journal*, 60(7), 102–121.

Wei, W., & Yang, L. L. (2025). The impact of artificial intelligence use on employee work performance: Based on the functional and relational perspectives. *Management Review*, 37(10), 162–173.

Yang, L., Zou, X. C., & Ma, Y. K. (2025). Strengthen financial innovation and empower the development of new quality productive forces. *Macroeconomic Management*, (6), 78–85.

Yang, Y., & Su, X. (2024). Spatial correlation network structure of carbon emission reduction capacity on urban agglomerations and its driving factors: A perspective of Sustainable Development Goals. *Sustainable Cities and Society*, 113, Article 105646. <https://doi.org/10.1016/j.scs.2024.105646>

Yang, Y., Liu, S. W., & Deng, X. Y. (2025). Risks and opportunities: A study on the dual effects of artificial intelligence on corporate resilience. *Nankai Business Review*, 1–39.

Ye, C. S., Li, Z. W., & Sun, W. (2022). How can the government and the market drive the innovation of technological SMEs through “two wheels”? Policy evaluation of “the pilot policy of combining science and technology with finance” based on micro-level enterprise data. *Inquiry into Economic Issues*, (5), 32–46.

Zhang, J. T., Wu, Y. Y., & Zhu, Y. (2025). Spatial correlation network and its driving mechanism of collaborative agglomeration between manufacturing and producer services in China. *Acta Geographica Sinica*, 80(2), 396–414. <https://doi.org/10.11821/dlxb202502009>

Zhou, K., Qu, Z., Zeng, S. P., & Li, Z. K. (2025). Sci-tech finance and key digital technological breakthroughs: Evidence from SRDI firms. *The Journal of World Economy*, 48(11), 3–32.

Zhou, L. H., Qi, X. F., & Li, M. F. (2024). Technology finance, supply chain spillovers, and labor employment: Evidence from pilot policies to promote the integration of technology and finance. *Economic Perspectives*, (12), 18–35.

Zhu, K., & Guo, L. (2024). Financial technology, inclusive finance and bank performance. *Finance Research Letters*, 60, Article 104872. <https://doi.org/10.1016/j.frl.2023.104872>

Zou, K., Liu, X., & Li, X. M. (2024). The generation effect and mechanism of new quality productivity in the development of technology finance. *Financial Economics Research*, 39(6), 3–18.

Author Introduction

First author, Huan Chen, School of International Business, Hainan University, email: chenhuan9422@163.com. The author's contributions include conceptualization, methodology, software, data curation, writing -original draft preparation.

Corresponding author, Muhan Liang, School of Information and Communication Engineering, Hainan University, email: 479332290@qq.com. The author's contribution involves supervision, writing- reviewing and editing.